Vehicle Visual Defect Detection

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THE UNIVERSITY OF CHICAGO x latentview
Actionable Insights • Accurate Decisions

LatentView Analytics – An Introduction

Organization Overview

- LatentView Analytics is a leading digital analytics firm founded in 2006
- The company specializes in blending digital and traditional data for decision optimization.
- Based in India, LatentView supports Fortune 500 clients across the U.S., Europe, and Asia.

As of 31st March 2023

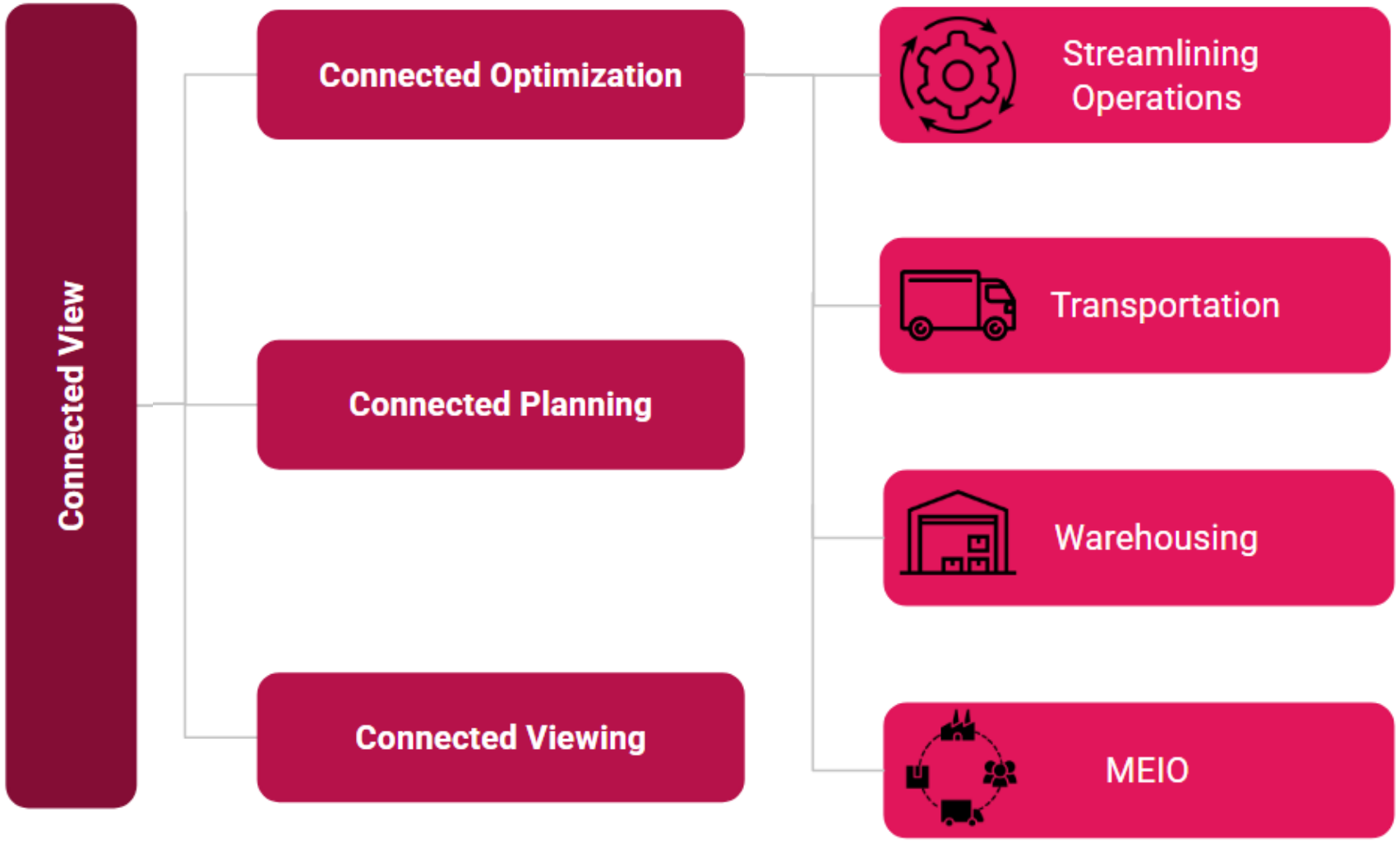
Awards & Recognition

- Great Place To Work. 2022-2023
- Microsoft ESOMAR Netherlands Conference 2017
- NASSCOM AI Award Winner 2018
- Analytics Solutions Provider of the Year 2017
- Gartner Market Guide for Data & Analytics Service Providers 2019
- Deloitte. Deloitte technology fast 50

Key Partnerships

- aws partner network | service delivery
- Optimizely
- Google Cloud Partner
- snowflake
- looker
- Microsoft Partner Gold Data Analytics
- tableau Tableau Alliance Partner Program
- Adobe SOLUTION PARTNER
- databricks

Proprietary Framework that provides a holistic, cross-functional view of business performance



This is where we come in!

Current Vehicle Inspection Process

[Tsutsumi Plant] Quality Control and Inspection - YouTube

Current Challenges in Vehicle Inspection

Current vehicle inspection workflow :

Vehicle completes production → Inspectors perform visual checks → Defects are tagged → Details manually entered into QC system → Vehicle sent for rework

The Business Problem:

Visual inspection of vehicles for defects is currently *manual, subjective, and inefficient*, especially for appearance-related issues such as paint scratches, dents, or surface irregularities.

New Opportunities from the AI Solution

AI Solution

- Utilize Computer Vision Techniques
- Implement an Automation system

Process Improvement

Car manufacturing with a
- *More efficient*
- *Reliable*
- *Scalable*
- *Cost-effective*

solution for defect inspections.

Expand Profile

- Expand the portfolio under the Supply Chain asset
- Support the *Connected View* framework

Proposed Solution

Current vehicle inspection workflow :

Vehicle completes production → Inspectors perform visual checks → Defects are tagged → Details manually entered into QC system → Vehicle sent for rework

New vehicle inspection workflow :

Vehicle completes production → CCTV records video of the vehicle → Model detects defects in real time → Defect details displayed on web interface → Vehicle sent for rework

Proposed Solution

Develop a scalable computer vision system that automatically processes vehicle videos to detect defects, improving accuracy, speed, and traceability in vehicle inspection processes.

An end-to-end AI application that:

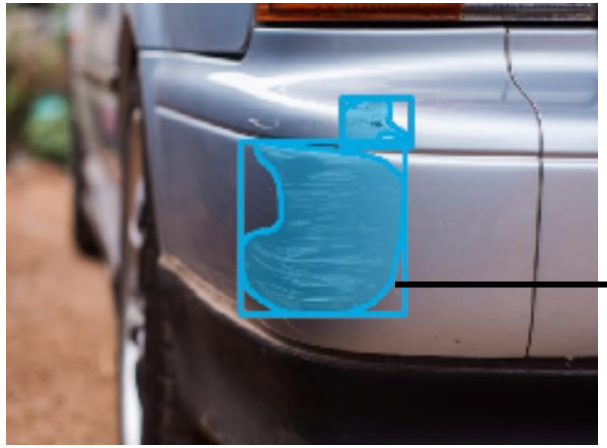
01 Accepts a video input of a vehicle

02 Splits it into frames

03 Detects visual defects

04 Outputs tagged results with job IDs

Example output:

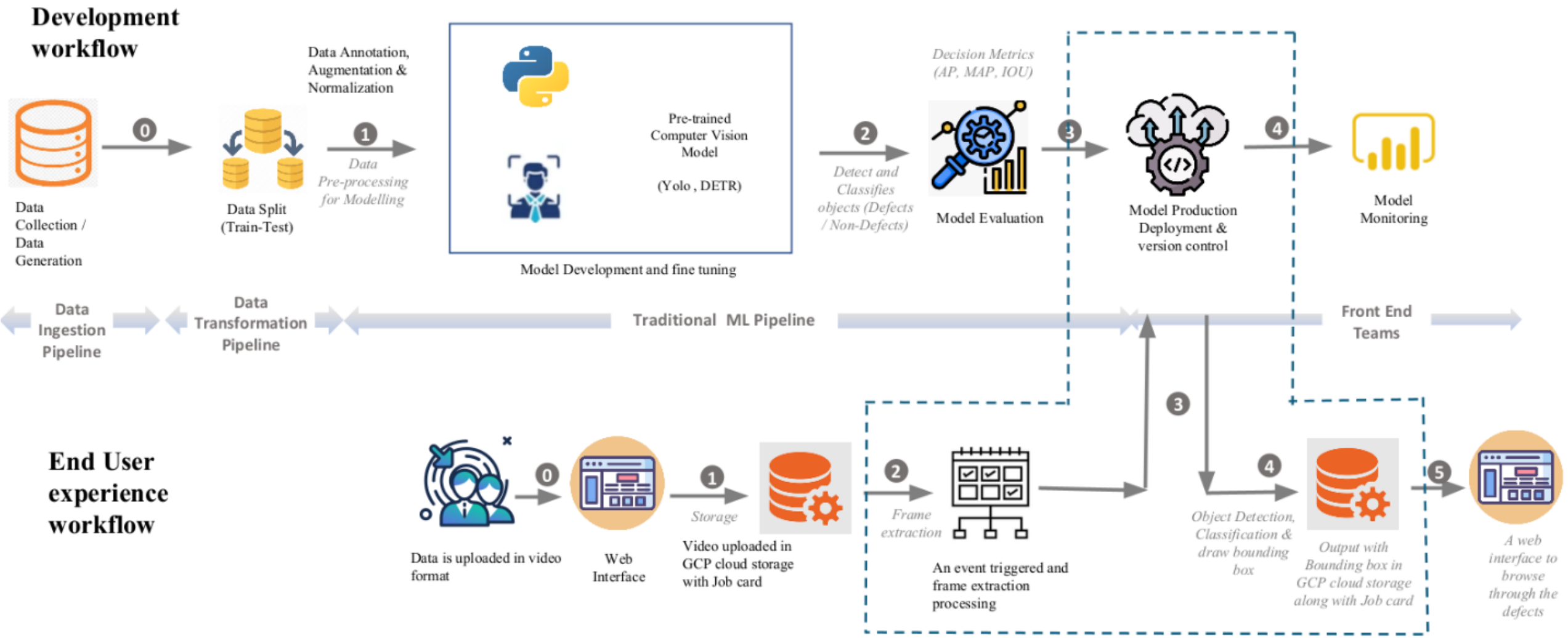


01 Type of defect

02 Location on vehicle

03 Job IDs

End-to-End Technical Pipeline (6 steps)



Literature Review Summary

Vehicle Damage Classification using Three Quarter View (2024):

https://www.sciencedirect.com/science/article/pii/S2405844024100473#se0080

- The authors present the TQVCD dataset, which comprises publicly available images of vehicles captured from a three-quarter perspective
- Used to output binary classification of Damaged vs. Undamaged vehicles
- Paper also contains information of currently updated vehicle damage datasets curated by their purpose (classification, object detection, segmentation)

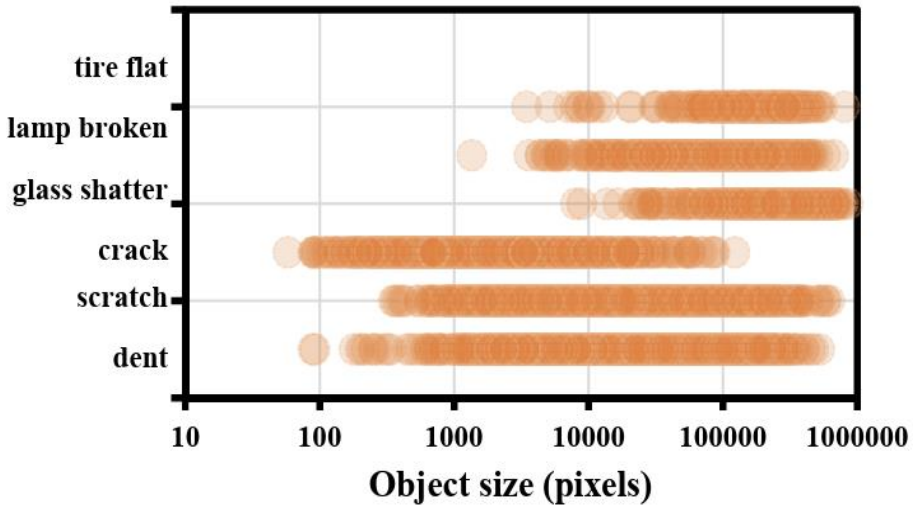
CarDD Dataset – a New Benchmark for Vision Based Car Damage Detection (2022):

https://arxiv.org/abs/2211.00945

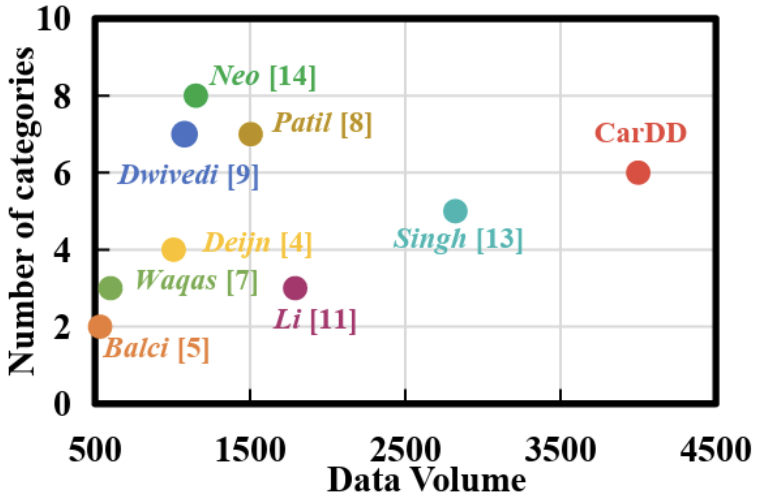
- The authors present the CarDD dataset, which is first publicly available large-scale dataset designed for vision-based car damage detection and segmentation
- Popular object identification models like YOLO and Mask R-CNN are used for testing this dataset
- Multi-class classification output identifying type of damage and bounding box identifying position of damage

CarDD – The Benchmark Dataset for Vehicle Defect Detection

Object size distribution



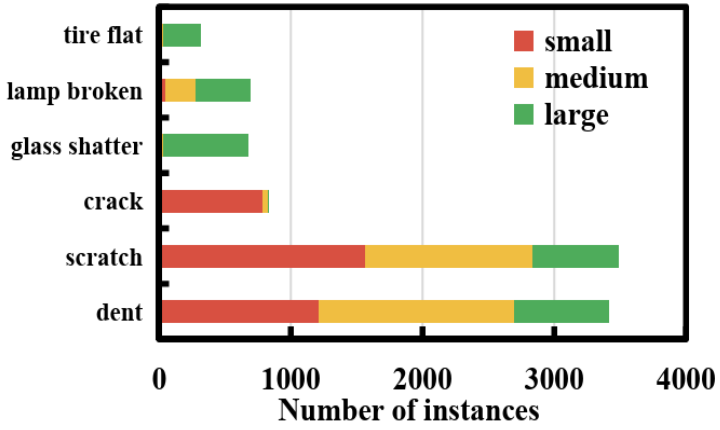
Comparison of datasets



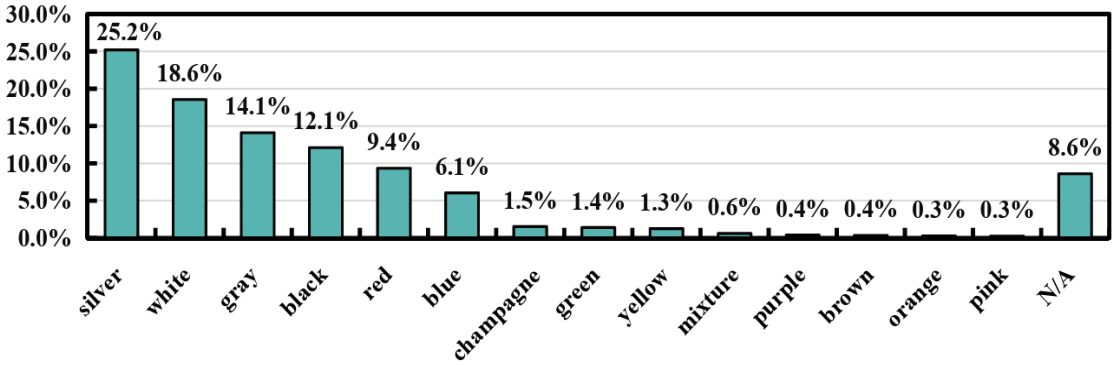
- Contains 4,000 labeled images of vehicle dents, scratches, and cracks.
- Average resolution of CarDD is 684,231 pixels, 13.6 times higher than other publicly available vehicle damage datasets.
- Has annotations to perform classification, object detection, instance segmentation, and salient object detection (SOD).

Data Gaps and Augmentations

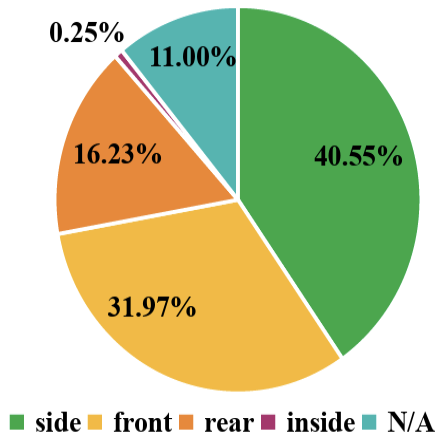
Instance scale distribution



Vehicle color distribution



Shooting angle distribution



Data Gaps

- Damage type imbalance: scratches and dents dominate.
- Angle and color imbalance: front-view bias; silver/white/black/red cars overrepresented.

Augmentation Plan

- Use color shifting, affine transforms and few- shot models. Synthetic data generation (GANs) is currently optional.
- Source and combine additional Datasets from TQVCD paper, Kaggle, Roboflow, and Shaip for better class diversity.

Preliminary Model Approach

- **Model Selection**

YOLOv11 with OBB (Oriented Bounding Box) support for better fit of slanted or rotated defects.

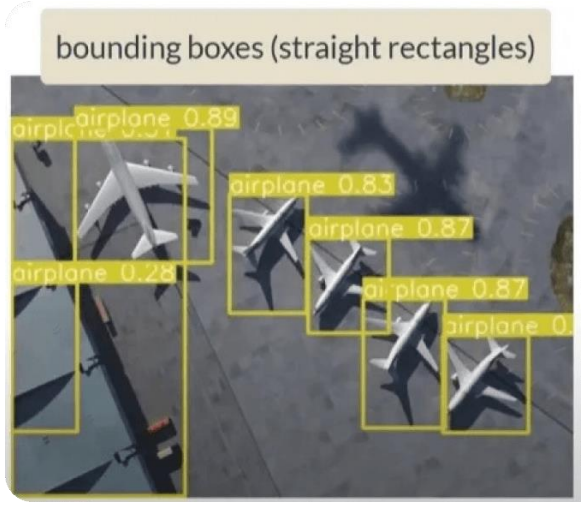
- **Pretrained Weights**

Using weights pretrained on COCO or defect-related tasks for improved accuracy.

- **Strategy behind YOLO11**

OBB reduces need for exhaustive rotation-based augmentation.

Focus on augmenting underrepresented defect types and lighting conditions.



Model Evaluation and Expected Finding

01

Defect Detection Model

- **Mean Average Precision (mAP@0.5:0.95 IoU threshold ≥ 0.80)** capable of detecting key defect types with visual clarity.
- Ability to:
  1. Identify and localize multiple defects per frame.
  2. Handle rotated or slanted views using OBB.
  3. Generalize across vehicle colors and angles.

02

Business Value

- Enhanced Inspection Throughput :
  1. Faster defect detection than manual methods.
  2. Standardized and traceable visual defect logging
- Integration Benefits :
  1. Reduce quality inspection times and improve customer satisfaction.
  2. Serve as a compelling sales accelerator demo for clients in automotive manufacturing or insurance sectors.

Questions?

Literature Review Citations

Lee, D., Lee, J., & Park, E. (2024). Automated vehicle damage classification using the three-quarter view car damage dataset and deep learning approaches. *Heliyon*, *10*(14), e34016. https://doi.org/10.1016/j.heliyon.2024.e34016

Wang, X., Li, W., & Wu, Z. (2023). CarDD: A New Dataset for Vision-Based Car Damage Detection. *IEEE Transactions on Intelligent Transportation Systems*, *24*(7), 7202–7214. https://doi.org/10.1109/tits.2023.3258480